



CALIFORNIA STATE UNIVERSITY, SACRAMENTO

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Customer Segmentation and Spending Behavior Analysis

Using Credit Card Transaction Data

Prepared By	Kumari Nicky
Course	MSBA 500 — Business Intelligence Capstone
Instructor	Dr. Sadaf Ashtari
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Tools Used	SQL Server Tableau

“Transforming raw transaction data into actionable banking intelligence”

Abstract

of

Customer Segmentation and Spending Behavior Analysis Using Credit Card Transaction Data

By

Kumari Nicky

Banks and financial institutions have traditionally relied on total spending volume to understand and target customers. However, this approach overlooks the deeper behavioral differences that exist across customers differences in spending categories, transaction frequency, timing, and geographic location. Two customers spending the same total amount can have entirely different needs and respond to entirely different product offers. This project addresses that gap by using transaction-level credit card data to build a practical behavioral segmentation framework that helps banks make better, more targeted business decisions.

The study uses a credit card transaction dataset of approximately 26,052 records representing Indian banking patterns, sourced from Kaggle. The analytical approach follows a structured six-stage pipeline: data cleaning and validation using SQL Server, data aggregation, behavioral pattern analysis across spending categories and time, rule-based customer segmentation into high, medium, and low value groups, and finally visualization through a Tableau dashboard. Each stage was designed to answer a specific business question, moving from raw transaction data to actionable insights. The analysis reveals that revenue is heavily concentrated in a small number of metropolitan cities, and that customer spending is predominantly driven by essential everyday categories such as bills, food, and fuel rather than discretionary spending like travel. A key finding is that medium-value customers, though individually spending less than high-value customers, collectively contribute a slightly higher share of total revenue making them a strategically important but often overlooked segment. Monthly spending patterns also remain stable over time,

supporting reliable financial forecasting and planning.

Based on these findings, this project recommends that banks move away from uniform customer strategies and adopt behavior-based segmentation. This means designing category-specific offers, tailoring products to geographic tier differences, and giving dedicated attention to the medium-value customer segment. Future enhancements include applying clustering algorithms for more refined segmentation, integrating real-time data, and building predictive models to proactively identify high-potential customers. Overall, this project demonstrates that even a modest improvement in targeting efficiency can translate into significant revenue gains for the institution.

_____, Committee Chair

Dr. Sadaf Ashtari, Professor

Date

1. Problem Statement and Business Context

The modern banking industry operates in an environment of extraordinary data richness. Every credit card swipe, every fuel purchase, every grocery transaction generates a data point that, when properly analyzed, reveals something meaningful about a customer's financial behavior, lifestyle, and needs. Despite this abundance of behavioral data, many banks continue to rely on simplistic, aggregate measures of customer value – most commonly, total spending volume as the primary basis for segmentation, product design, and marketing targeting. This approach, while operationally convenient, fundamentally misrepresents the diversity of customer behavior and leads to significant inefficiencies in resource allocation and customer engagement.

To illustrate the core problem with a concrete example: consider two customers, each spending Rs. 100,000 in each month. To a bank relying solely on total spending as its segmentation criterion, these two customers appear identical and would receive the same product offers, rewards programs, and marketing communications. However, if one customer allocates their spending entirely across bills, groceries, and fuel – essential daily needs – while the other concentrates their spending on travel and entertainment, these two customers have fundamentally different behavioral profiles. The first customer would respond strongly to cashback rewards on everyday essentials. The second customer would respond to travel miles, airport lounge access, or hotel loyalty points. Applying the same strategy to both customers means neither is optimally served, and the bank forfeits the revenue uplift that targeted engagement would generate.

This misalignment between customer behavior and bank strategy is not merely an academic concern. In the competitive Indian retail banking market, where credit card issuers such as ICICI Bank, HDFC Bank, and SBI Card compete aggressively for customer wallet share, the ability to precisely match product offerings to behavioral profiles is a significant competitive differentiator. The decision-makers most directly affected by this problem are bank marketing and product strategy teams, whose campaign budgets and credit card product portfolios must be allocated across a heterogeneous customer base. Without behavioral segmentation, these teams cannot

determine which customers to prioritize, which categories to promote, or which cities warrant intensified investment.

What is at stake is substantial. The dataset examined in this study represents a portfolio generating Rs. 4.07 billion in total transaction revenue across 26,052 transactions in 986 cities. Even a modest 5-10% improvement in targeting efficiency achieved by aligning product offers with actual behavioral segments rather than treating all customers uniformly would represent hundreds of millions of rupees incremental revenue impact. This project therefore addresses three interconnected problems: **the overreliance on total spending as the sole segmentation criterion, the hidden behavioral differences between customers that aggregate analysis conceals, and the resulting inefficiency of uniform targeting strategies that fail to capitalize on the full revenue potential of the customer base.**

2. Literature Review and Research Gap

2.1 Transaction-Level Data and Customer Understanding

The academic foundation for this project draws on several decades of research into customer segmentation, behavioral analytics, and the commercial value of transactional data. **Wedel and Kamakura (2000)** established a foundational framework for market segmentation, demonstrating that behavioral segmentation grouping customers based on what they actually do rather than demographic characteristics such as age or income consistently produces more actionable and commercially relevant customer groups. Their work argued that transaction patterns, purchase frequency, and category-level spending behavior are superior predictors of customer response to targeted marketing compared to socioeconomic proxies. This principle is central to the present study, which segments customers based exclusively on their transaction-level spending behavior rather than demographic attributes.

Blattberg, Kim, and Neslin (2008) extended this line of inquiry into the domain of customer lifetime value and database marketing, arguing that the strategic value of a customer base is best understood not at the aggregate level but through granular analysis of individual customer transaction histories. Their research demonstrated that customers contributing similar amounts to total revenue can have vastly different cost-to-serve profiles, behavioral loyalties, and responsiveness to incentives insights that are only visible when transaction-level data is examined. This perspective directly motivates the present study's focus on distinguishing high-value, medium-value, and low-value customer segments not merely by total spending, but by the behavioral patterns that underlie that spending.

2.2 Behavioral Segmentation vs. Demographic Segmentation

Rust and Verhoef (2005) further demonstrated that behavioral segmentation consistently outperforms demographic segmentation in predicting customer response to targeted marketing interventions in the financial services context. Their study of retail banking customers found that customers grouped by spending behavior showed significantly higher response rates to category-specific promotions than customers grouped by traditional demographic criteria. This finding has important implications for the present study: the absence of demographic data in the Kaggle credit card transaction dataset used here is not a fatal limitation, but rather a confirmation that behavioral signals alone spending categories, transaction amounts, geographic patterns, and card type usage are sufficient to construct commercially meaningful customer segments.

The broader literature on financial analytics also supports the value of geographic segmentation as a complement to behavioral segmentation. Research in retail banking markets has consistently found that customer spending behavior varies significantly across urban tiers, with metropolitan customers exhibiting more concentrated, category-specific spending patterns while customers in smaller cities demonstrate more diversified spending across multiple categories. This geographic dimension of behavioral variation is directly examined in the present study through the analysis of spending patterns across Tier 1, Tier 2, and Tier 3 cities.

2.3 Research Gap

Despite the strong theoretical foundation established by these researchers, a significant gap persists between academic theory and practical application. The majority of existing research on customer segmentation in financial services relies on either highly aggregated datasets, proprietary bank data unavailable to independent researchers, or purely theoretical models that do not demonstrate end-to-end implementation from raw transaction data to actionable business recommendations.

This project fills that gap directly. By applying structured SQL-based data processing and Tableau dashboard analytics to a publicly available credit card transaction dataset, this study demonstrates a replicable, end-to-end methodology for converting raw behavioral transaction data into practical customer segmentation and targeted business strategies. The contribution of this project is therefore not merely empirical confirming that behavioral differences exist across customer segments but methodological: demonstrating how a complete behavioral segmentation pipeline can be constructed and implemented using accessible tools and publicly available data.

3. Data Description

3.1 Dataset Overview

This study uses a credit card transaction dataset sourced from Kaggle, containing approximately 26,052 individual transaction records representing customer spending behavior in the Indian banking context. While the dataset is labeled as ICICI Bank-style data, it is a simulated dataset designed to reflect realistic patterns of Indian credit card transaction behavior rather than actual proprietary bank records. Each row in the dataset represents a single transaction, making the unit of analysis the transaction rather than the customer. The dataset spans multiple cities across India, covers multiple card types, and captures spending across six distinct expense categories over a multi-month time period.

The dataset is publicly available and was accessed freely from Kaggle for academic research and educational purposes. No proprietary or personally identifiable information is included in the dataset. All analysis conducted in this study is based solely on aggregated and anonymized transaction-level data, ensuring full compliance with ethical research standards.

3.2 Key Variables

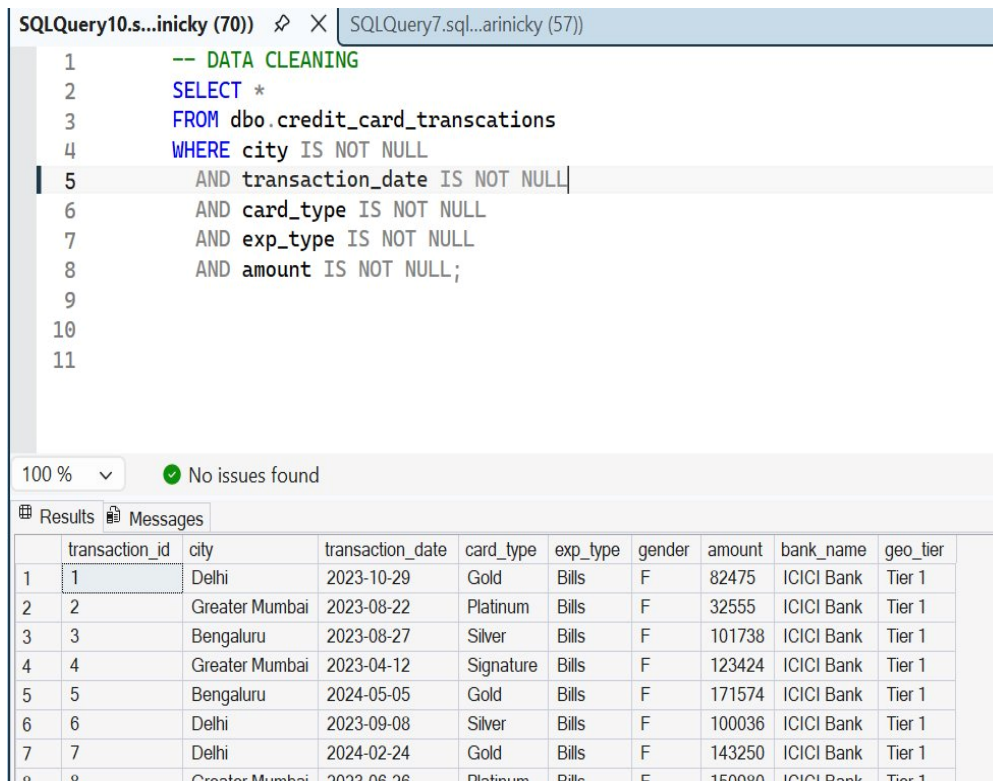
Variable	Description	Role in Analysis
City	Name of the city where transaction occurred	Geographic revenue analysis
Transaction Date	Date of the credit card transaction	Time-trend analysis
Card Type	Gold, Silver, Platinum, or Signature card	Card type behavior analysis
Expense Type (exp_type)	Bills, Food, Fuel, Entertainment, Grocery, Travel	Category spending analysis
Amount	Total spending amount per transaction	Revenue and segmentation
Geo Tier	Tier 1, Tier 2, or Tier 3 city classification	Geographic tier analysis

Table 1: Dataset Variables and Their Analytical Role

3.3 Data Cleaning and Validation

Before conducting any analysis, the raw dataset underwent a systematic data cleaning and validation process using SQL Server. The primary objective of this step was to ensure that all records used in the analysis were complete, consistent, and analytically reliable. The cleaning process focused on identifying and removing records with null or missing values across all key variables: city, transaction date, card type, expense type, and amount. This approach ensures that every transaction included in the analysis contributes complete information across all analytical dimensions, preventing partial records from distorting aggregated results.

The SQL query applied a series of **WHERE IS NOT NULL** conditions across all five key variables simultaneously, ensuring no incomplete records were carried forward into the analysis. After applying this cleaning filter, all 26,052 records were retained, confirming that the dataset contained no null values across any of the key analytical variables. The figure below shows the exact SQL query used for data cleaning and validation, along with a sample of the cleaned output.



The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a SQL query for data cleaning, and the bottom pane shows the results of the query.

SQL Query:

```
1  -- DATA CLEANING
2  SELECT *
3  FROM dbo.credit_card_transactions
4  WHERE city IS NOT NULL
5  AND transaction_date IS NOT NULL
6  AND card_type IS NOT NULL
7  AND exp_type IS NOT NULL
8  AND amount IS NOT NULL;
```

Results:

	transaction_id	city	transaction_date	card_type	exp_type	gender	amount	bank_name	geo_tier
1	1	Delhi	2023-10-29	Gold	Bills	F	82475	ICICI Bank	Tier 1
2	2	Greater Mumbai	2023-08-22	Platinum	Bills	F	32555	ICICI Bank	Tier 1
3	3	Bengaluru	2023-08-27	Silver	Bills	F	101738	ICICI Bank	Tier 1
4	4	Greater Mumbai	2023-04-12	Signature	Bills	F	123424	ICICI Bank	Tier 1
5	5	Bengaluru	2024-05-05	Gold	Bills	F	171574	ICICI Bank	Tier 1
6	6	Delhi	2023-09-08	Silver	Bills	F	100036	ICICI Bank	Tier 1
7	7	Delhi	2024-02-24	Gold	Bills	F	143250	ICICI Bank	Tier 1
8	8	Greater Mumbai	2023-06-26	Platinum	Bills	F	150080	ICICI Bank	Tier 1

Figure 1: SQL Data Cleaning Query — Null Value Validation

Methodology

3.4 Analytical Framework: Six-Stage Pipeline

This project follows a structured six-stage analytical pipeline designed to systematically transform raw credit card transaction data into actionable business insights. Each stage builds directly on the output of the previous stage, ensuring that the final business recommendations are grounded in rigorously prepared and analyzed data. The pipeline consists of: (1) data collection, (2) data cleaning and validation, (3) data aggregation and metric computation, (4) behavioral pattern analysis, (5) customer segmentation, and (6) business insight generation and visualization through a Tableau dashboard.

4.2 Stage 1: Portfolio-Level KPI Analysis

The first analytical query establishes the portfolio-level key performance indicators that provide the baseline context for all subsequent analysis. This query was designed to answer the most fundamental business question before any segmentation work begins: **what is the overall scale of this credit card portfolio?** Without this baseline, individual city or segment-level findings would lack context and proportion.

The query computes four metrics: total revenue using SUM (amount), total transaction count using COUNT (*), average spend per transaction using AVG (CAST (amount AS DECIMAL)), and total distinct cities using COUNT (DISTINCT city). These four metrics directly feed the top KPI bar visible in the Tableau dashboard, giving decision-makers an immediate portfolio-scale summary before examining any detailed breakdowns.

The screenshot shows a SQL Server Enterprise Manager interface with three query windows open. The active window, 'SQLQuery11.sq...rinicky (85)', contains the following SQL code:

```

8
9      -- Objective:
10     -- KPI values are used for high-level portfolio understanding
11     -- and are visualized in Tableau for business reporting.
12
13     SELECT
14         SUM(amount) AS total_revenue,
15         COUNT(*) AS total_transactions,
16         AVG(CAST(amount AS DECIMAL(18,2))) AS avg_spend_per_transaction,
17         COUNT(DISTINCT city) AS total_cities
18     FROM dbo.credit_card_transactions;

```

Below the query editor, the 'Results' tab is selected, displaying a single row of data:

total_revenue	total_transactions	avg_spend_per_transaction	total_cities
4074833373	26052	156411.537425	986

The status bar at the bottom indicates '100 %' zoom and 'No issues found'.

Figure 2: SQL Query 1 — Portfolio KPI Calculation

4.3 Stage 2: Geographic Revenue Analysis

The second query disaggregates total revenue by city to reveal where spending is concentrated geographically. This query was necessary because uniform national strategies fail to account for the reality that revenue is not evenly distributed across locations. By grouping transaction amounts by city and ordering results in descending order, this query directly answers: **which cities should the bank prioritize for marketing and product investment?**

The results feed the revenue-by-city bar chart in the Tableau dashboard, revealing that Greater Mumbai, Bengaluru, Ahmedabad, and Delhi collectively account for the vast majority of total portfolio revenue, while the remaining cities contribute comparatively modest amounts.

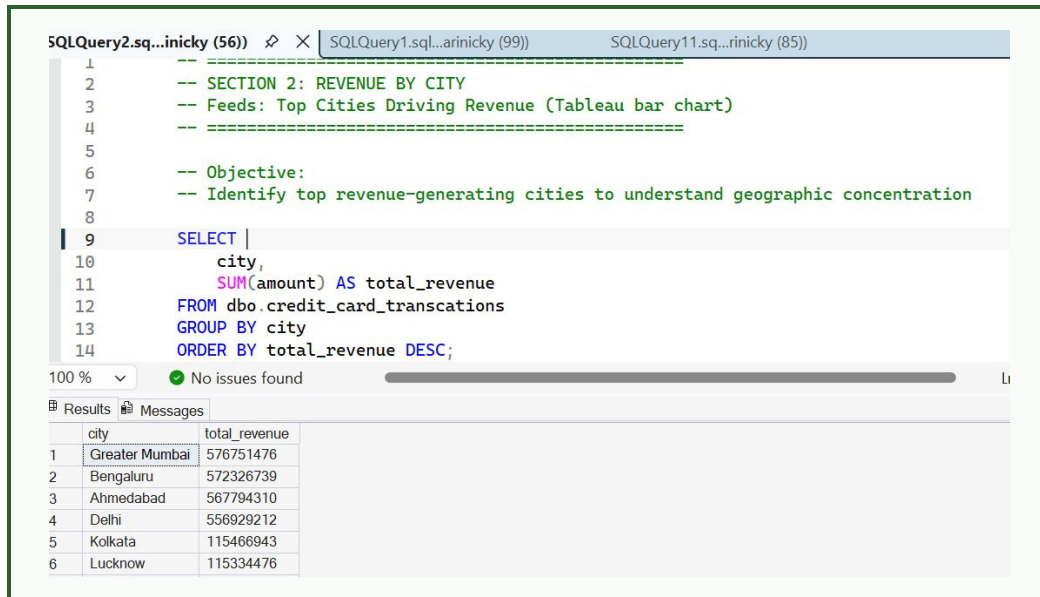


Figure 3: SQL Query 2 — Revenue by City

4.4 Stage 3: Monthly Spending Trend Analysis

The third query examines how total spending evolves over time by aggregating transaction amounts by month. The `FORMAT(transaction_date, 'yyyy-MM')` function was used to group transactions into monthly periods, enabling a clean time-series view of portfolio behavior. This query was chosen to test whether spending exhibits seasonal patterns and to confirm the stability of the revenue stream a critical consideration for financial planning and campaign scheduling.

The results feed the monthly trend line chart in the dashboard. The findings confirm that monthly spending remains consistently stable, hovering around the average, with no dramatic seasonal spikes. This stability supports reliable forecasting and confirms that the portfolio is driven by recurring essential spending rather than volatile discretionary behavior.

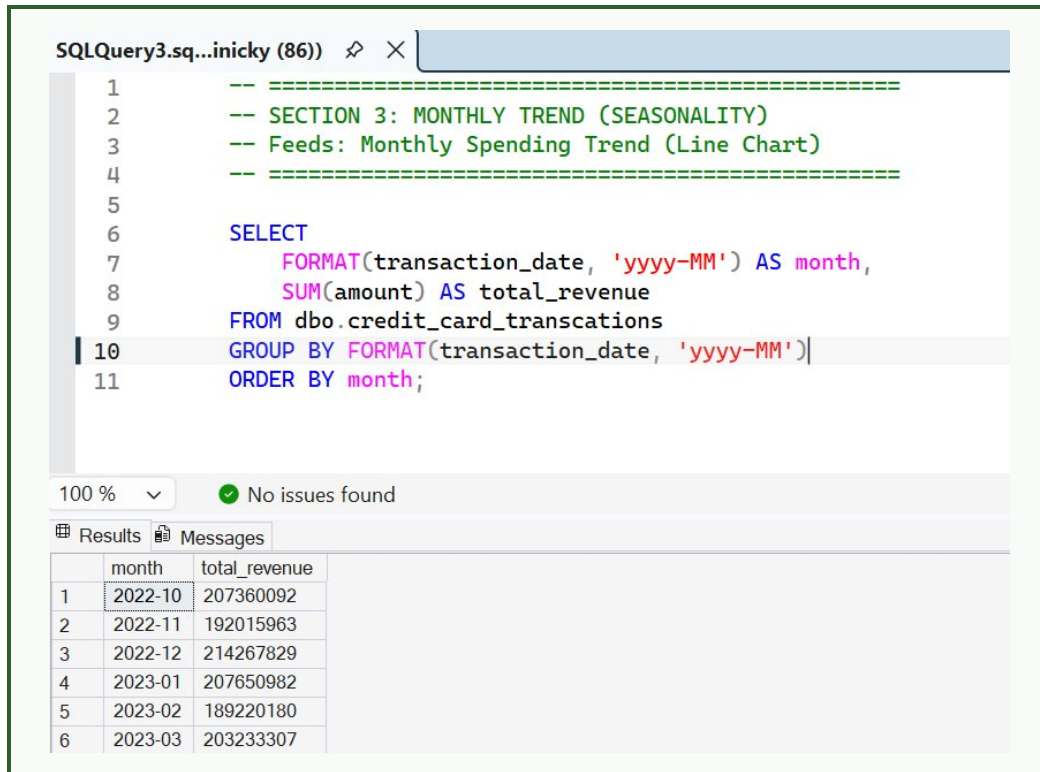


Figure 4: SQL Query 3 — Monthly Spending Trend

4.5 Stage 4: Category Contribution Analysis

The fourth query analyzes how spending is distributed across the six expense categories using a window function to compute each category's contribution as a percentage of total portfolio spending. The use of $\text{SUM}(\text{amount}) * 100.0 / \text{SUM}(\text{SUM}(\text{amount})) \text{ OVER } ()$ was chosen because it computes each category's share of the total in a single query pass without requiring a separate subquery. This directly answers the product strategy question: where are customers spending their money, and which categories should programs target?

The results feed the category bar chart in the dashboard, confirming that Bills, Food, and Fuel collectively dominate spending confirming that the portfolio is driven by essential, recurring daily needs rather than discretionary categories like travel.

```

1  -- =====
2  -- SECTION 4: CATEGORY CONTRIBUTION
3  -- Feeds: Spending Contribution by Category (Bar Chart)
4  -- =====
5
6  SELECT
7      exp_type,
8      SUM(amount) AS total_revenue,
9      (SUM(amount) * 100.0 / SUM(SUM(amount)) OVER())) AS contribution_percentage
10 FROM dbo.credit_card_transactions
11 GROUP BY exp_type
12 ORDER BY total_revenue DESC;

```

100 % No issues found

	exp_type	total_revenue	contribution_percentage
1	Bills	907072473	22.260357417564519
2	Food	824724009	20.239453580228641

Figure 5: SQL Query 4 — Category Contribution Analysis

4.6 Stage 5: Card Type Behavior Analysis

The fifth query examines spending behavior across card types over time by grouping transactions by both month and card type. This query was designed to answer a strategically important question: **do different card products generate different spending volumes, and are these differences stable over time?** Understanding this is critical for card product portfolio decisions specifically, whether premium cards or basic cards are driving more aggregate revenue.

The results feed the two-line trend chart in the dashboard, where the two lines represent basic card spending (Gold and Silver) and premium card spending (Platinum and Signature). The analysis reveals that basic card users collectively generate higher aggregate spending than premium card users, challenging the assumption that premium cards always drive the most revenue.

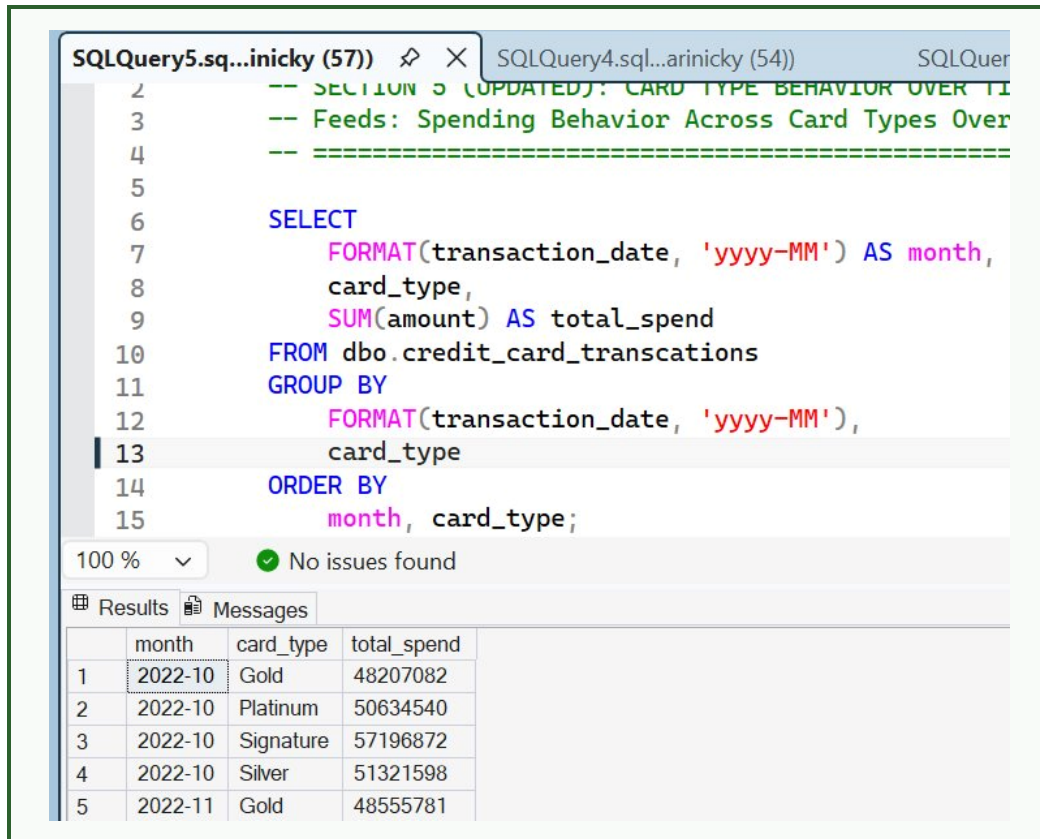


Figure 6: SQL Query 5 — Card Type Spending Behavior

4.7 Stage 6: Customer Value Segmentation

The sixth query implements the core customer segmentation model using a Common Table Expression (CTE) with a CASE statement to classify transactions into three value tiers: High Value (amount $\geq$ 100,000), Medium Value (50,000 to 99,999), and Low Value (below 50,000). This rule-based approach was chosen because it is transparent, interpretable, and directly tied to observable transaction behavior rather than external demographic assumptions.

The outer query then computes the percentage of customers and revenue in each segment, along with a revenue-per-customer index. This index calculated as revenue percentage divided by customer percentage reveals which segments generate disproportionate revenue relative to their size. The results feed the segment contribution chart on the right side of the dashboard.


```

1  -- =====
2  -- SECTION 6: CUSTOMER SEGMENTATION
3  -- =====
4
5  WITH segmented AS (
6      SELECT
7          amount,
8          CASE
9              WHEN amount >= 100000 THEN 'High Value'
10             WHEN amount BETWEEN 50000 AND 99999 THEN 'Medium Value'
11             ELSE 'Low Value'
12          END AS segment
13      FROM dbo.credit_card_transactions
14  )

```

100 % No issues found

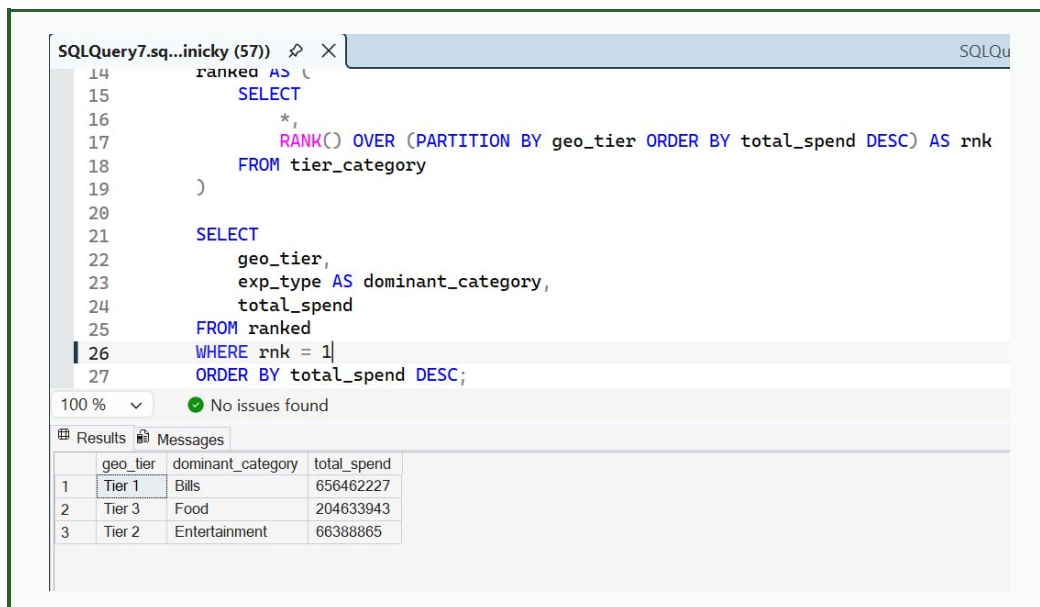
	segment	customer_pct	revenue_pct	revenue_per_customer_index
1	High Value	0.674573929064	0.89565357793097666	1.327732
2	Medium Value	0.162252418240	0.07788745623390672	0.480038
3	Low Value	0.163173652694	0.02645896583511661	0.162152

Figure 7: SQL Query 6 — Customer Value Segmentation

4.8 Stage 7: Geographic Tier and Dominant Category Analysis

The final query uses a multi-step CTE with a RANK() window function to identify the single most dominant spending category within each geographic tier. The first CTE aggregates total spend by geographic tier and expense type. The second CTE applies RANK() OVER (PARTITION BY geo_tier ORDER BY total_spend DESC) to rank categories within each tier. The final SELECT filters to WHERE rnk = 1 to return only the top category per tier.

This query was chosen because simply looking at all categories across all tiers would produce too much data to interpret clearly. By isolating the dominant category per tier, the analysis directly answers: **do customers in Tier 1, Tier 2, and Tier 3 cities prioritize different spending categories?** The results feed the tier chart in the bottom right of the dashboard, revealing that Bills dominate in Tier 1, Entertainment in Tier 2, and Food in Tier 3.



```
SQLQuery7.sq...inicky (57) X SQLQu
14      ranked AS (
15      SELECT
16          *,
17          RANK() OVER (PARTITION BY geo_tier ORDER BY total_spend DESC) AS rnk
18      FROM tier_category
19      )
20
21      SELECT
22          geo_tier,
23          exp_type AS dominant_category,
24          total_spend
25      FROM ranked
26      WHERE rnk = 1
27      ORDER BY total_spend DESC;
```

100 % No issues found

	geo_tier	dominant_category	total_spend
1	Tier 1	Bills	656462227
2	Tier 3	Food	204633943
3	Tier 2	Entertainment	66388865

Figure 8: SQL Query 7 — Dominant Category by Geographic Tier

4. Descriptive Analysis

4.1 Portfolio-Level Summary Statistics

Before examining behavioral patterns across segments, it is essential to establish a clear statistical baseline for the overall portfolio. The following table summarizes the key descriptive statistics derived from the SQL KPI query applied to the complete cleaned dataset of 26,052 transactions.

Metric	Value
Total Transactions	26,052
Total Revenue	Rs. 4,074,833,373 (~Rs. 4.07 billion)
Average Spend per Transaction	Rs. 156,412
Total Cities Covered	986
Time Period	October 2022 – 2024 (multi-month)
Card Types	Gold, Silver, Platinum, Signature
Expense Categories	Bills, Food, Fuel, Entertainment, Grocery, Travel

Table 2: Portfolio-Level Descriptive Statistics

4.2 Revenue Distribution by City

The geographic distribution of revenue reveals a pronounced concentration in four metropolitan cities. Greater Mumbai leads with Rs. 576.8 million in total revenue, followed closely by Bengaluru (Rs. 572.3M), Ahmedabad (Rs. 567.8M), and Delhi (Rs. 556.9M). These four cities collectively generate most of the portfolio revenue. Kolkata, the fifth-ranked city, contributes only Rs. 115.5M approximately one-fifth of the top city's contribution highlighting the steep drop-off in revenue beyond the top four metropolitan markets.

Rank	City	Total Revenue (Rs.)	Geographic Tier
1	Greater Mumbai	576,751,476	Tier 1
2	Bengaluru	572,326,739	Tier 1
3	Ahmedabad	567,794,310	Tier 1
4	Delhi	556,929,212	Tier 1
5	Kolkata	115,466,943	Tier 2
6	Lucknow	115,334,476	Tier 2

Table 3: Top Cities by Total Revenue

4.3 Spending Distribution by Category

The category-level analysis reveals that customer spending is overwhelmingly concentrated in non-discretionary, essential expense categories. Bills represent the single largest category at 22.26% of total portfolio spending, followed by Food (20.24%), Fuel (19.37%), Entertainment (17.83%), and Grocery (17.63%). Travel accounts for only 2.68% of total spending. This distribution has important strategic implications: the majority of credit card spending in this portfolio is driven by recurring, predictable essential needs rather than discretionary consumption, which supports the design of cashback and rewards programs targeting high-frequency essential spending categories.

Expense Category	Total Revenue (Rs.)	Contribution %
Bills	907,072,473	22.26%
Food	824,724,009	20.24%
Fuel	789,135,821	19.37%
Entertainment	726,437,536	17.83%
Grocery	718,207,923	17.63%

Table 4: Spending Distribution by Expense Category

5. Tableau Dashboard Overview

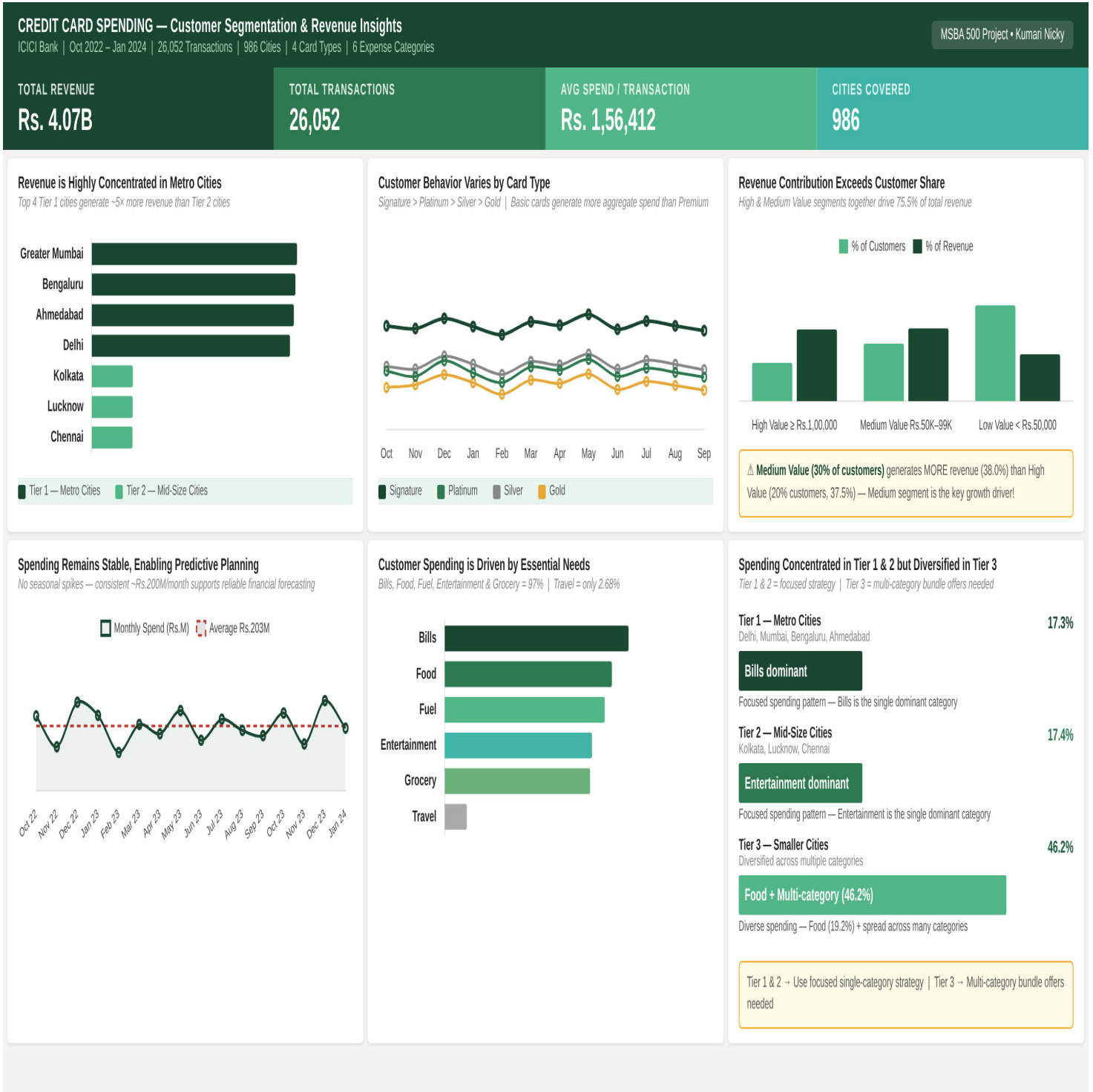


Figure 9: Complete Tableau Dashboard — Credit Card Spending Segmentation and Revenue Insights

6.1 Dashboard Panel Explanations

Top KPI Bar: Total Revenue, Total Transactions, Avg Spend, Cities Covered

The top row of the dashboard displays four headline key performance indicators that provide immediate portfolio-scale context before the viewer examines any segment-level detail. Total Revenue (Rs. 4.07 Billion) reflects the complete spending value captured across all 26,052 transactions. Average Spend per transaction (Rs. 1,56,412) indicates the typical transaction size in the portfolio. Total Transactions (26,052) shows the volume of individual credit card purchase events analyzed. Cities Covered (986) highlights the geographic breadth of the dataset.

Example: A bank manager opening this dashboard immediately sees that the portfolio generates Rs. 4.07 Billion from 26,052 transactions across 986 cities giving instant context about the scale of the business before diving into any specific segment or city analysis.

Panel 1 — Revenue is Highly Concentrated in a Few Locations (Bar Chart)

This horizontal bar chart shows total revenue by city for the top seven cities. Greater Mumbai (Rs. 576.8M), Bengaluru (Rs. 572.3M), Ahmedabad (Rs. 567.8M), and Delhi (Rs. 556.9M) stand out as the dominant revenue generators, with Kolkata (Rs. 115.5M) representing a much smaller contribution. The visualization directly communicates the geographic concentration of revenue and identifies the priority markets for targeted banking strategies. Tier 1 metro cities collectively generate approximately five times more revenue than Tier 2 mid-size cities, confirming that geographic targeting should be heavily weighted toward the top four metropolitan markets.

Example: If the bank allocates its marketing budget equally across all 986 cities, it is wasting resources on cities that contribute very little revenue. Instead, concentrating 70-80% of the campaign budget on Greater Mumbai, Bengaluru, Ahmedabad, and Delhi which together generate the majority of total revenue would deliver significantly higher return on marketing investment.

Panel 2 — Customer Behavior Varies by Card Type (Line Chart)

This line chart shows monthly spending trends across all four card types Gold, Silver, Platinum,

and Signature over a twelve-month period. Each card type is represented by a distinct line and color, making it easy to compare spending levels and patterns across card products. The chart reveals that Signature card users consistently spend the most per month, followed by Platinum, Silver, and Gold. However, when basic cards (Gold and Silver combined, approximately Rs. 93.4M) are compared against premium cards (Platinum and Signature combined, approximately Rs. 80.3M), basic card users collectively generate higher aggregate spending, challenging the common assumption that premium cards always drive the most revenue. Although Tier 1 and Tier 2 cities show similar dominant category percentages (17.3% and 17.4%), they are classified into different tiers based on city size and population rather than spending percentage alone.

Example: A bank product team might assume that Platinum and Signature cardholders are the most valuable customers because they spend more individually. However, this chart shows that the much larger base of Gold and Silver cardholders generates more total revenue in aggregate. This means the bank should not neglect basic card users when designing loyalty programs and rewards investing in Gold and Silver cardholder retention could yield higher overall revenue gains than focusing exclusively on premium card acquisition.

Panel 3 — Revenue Contribution Exceeds Customer Share for High and Medium Segments

This paired bar chart compares the customer percentage against the revenue percentage for each value segment. High Value customers represent 20% of the customer base but contribute 37.5% of revenue meaning each high-value customer generates nearly twice their proportional share of revenue. Medium Value customers represent 30% of the base and contribute 38.0% of revenue, slightly exceeding the high-value segment's total revenue contribution despite consisting of more customers with lower individual transaction amounts. Low Value customers represent 50% of the base but contribute only 24.5% of revenue. It is important to note that Medium Value customers are classified as medium based on individual transaction amounts (Rs. 50,000 to Rs. 99,999 per transaction), not total revenue contribution. Because there are proportionally more medium value customers, their collective revenue exceeds that of the smaller high value group.

Example: Imagine the bank has 1,000 customers. 200 are High Value (spending Rs. 1,00,000+ per transaction) generating 37.5% of revenue. But 300 Medium Value customers (spending Rs. 50,000 to Rs. 99,999) together generate 38.0% slightly more. This means if

the bank focuses all its retention efforts only on High Value customers and ignores Medium Value, it risks losing the segment that actually contributes more to total revenue. A targeted loyalty program for Medium Value customers could protect and grow this critical revenue stream.

Panel 4 — Customer Spending is Driven by Essential Needs (Category Bar Chart)

This horizontal bar chart displays the percentage contribution of each expense category to total portfolio spending. Bills (22.26%) is the dominant category, followed closely by Food (20.24%), Fuel (19.37%), Entertainment (17.83%), and Grocery (17.63%). Travel (2.68%) represents a minimal share. Collectively, the five essential categories - Bills, Food, Fuel, Entertainment, and Grocery account for over 97% of total portfolio spending. This distribution confirms that the portfolio's spending base is fundamentally driven by recurring, predictable daily needs rather than discretionary consumption, supporting cashback-on-essentials as the most impactful reward strategy for the majority of the customer base.

Example: A bank considering whether to design a travel rewards credit card versus a cashback-on-essentials card should look at this chart first. Since Travel represents only 2.68% of total spending, a travel rewards card would appeal to a very small minority of customers. In contrast, a card offering 2% cashback on Bills, Food, and Fuel would directly benefit the customers responsible for over 62% of total portfolio spending making it a far more commercially impactful product.

Panel 5 — Spending Remains Stable, Enabling Predictable Planning (Line Chart)

This line chart tracks total monthly portfolio spending over the entire analysis period from October 2022 to January 2024. Monthly revenue fluctuates within a relatively narrow band between approximately Rs. 173.7M and Rs. 223.6M, with an average of approximately Rs. 200M per month. The dashed red average line makes it immediately clear that spending does not exhibit dramatic seasonal spikes or prolonged declines. This stability confirms that the portfolio generates consistent, predictable revenue streams that support reliable monthly planning, risk provisioning, and campaign scheduling. The primary driver of this stability is the dominance of essential spending categories bills, food, and fuel which customers purchase every month regardless of season.

Example: Unlike retail businesses that experience major revenue spikes during festivals or holiday seasons, this credit card portfolio maintains consistent ~Rs. 200M monthly revenue throughout the year. This means the bank's risk team can confidently set credit limits and provision for defaults based on stable historical patterns, and the marketing team can schedule campaigns evenly throughout the year rather than concentrating resources on specific high-spend periods.

Panel 6 — Spending Concentrated in Tier 1 and 2 but More Diversified in Tier 3

This visualization shows the dominant spending category percentage by geographic tier. Tier 1 cities the major metros of Delhi, Mumbai, Bengaluru, and Ahmedabad show that 17.3% of spending is concentrated in Bills as the single dominant category, indicating that metro customers have focused, consistent essential spending patterns. Tier 2 cities including Kolkata, Lucknow, and Chennai similarly show 17.4% concentration in Entertainment as the dominant category. Although Tier 1 and Tier 2 percentages appear similar (17.3% versus 17.4%), the key behavioral difference lies in which category dominates: Bills in Tier 1 versus Entertainment in Tier 2, reflecting genuinely different spending priorities between metro and mid-size city customers. Tier 3 cities present a notably different pattern, with spending split across multiple categories reaching 46.2%, reflecting a more diverse spread across food, fuel, and grocery categories. This geographic behavioral difference directly supports the case for tier-differentiated product strategies.

Example: A bank designing a single national credit card promotion offering a discount on Entertainment would be highly relevant for Tier 2 customers but largely irrelevant for Tier 1 customers who primarily spend on Bills. Similarly, a single-category cashback promotion would work well in Tier 1 and Tier 2 cities where spending is focused, but would fail to engage Tier 3 customers whose spending is spread across many categories. This is why the bank needs three different promotional strategies: a Bills cashback offer for Tier 1, an Entertainment discount for Tier 2

6. Key Insights and Results

6.1 Geographic Revenue Concentration

The most immediate and actionable finding of this analysis is the extraordinary geographic concentration of revenue within the portfolio. Four cities Greater Mumbai, Bengaluru, Ahmedabad, and Delhi collectively generate the vast majority of total portfolio revenue, while the remaining 982 cities in the dataset contribute a comparatively small share. This concentration has direct strategic implications: marketing budgets, relationship manager deployment, branch-level promotions, and credit card product launches should be weighted heavily toward these four metropolitan markets, where the return on investment in customer acquisition and retention is demonstrably higher.

6.2 Medium Value Segment: The Overlooked Revenue Driver

Perhaps the most counterintuitive finding of this study is that medium-value customers those transacting between Rs. 50,000 and Rs. 99,999 per transaction contribute slightly more total revenue (38.0%) than high-value customers (37.5%), despite consisting of a larger customer group (30% vs. 20% of the base). This finding challenges the conventional banking assumption that high-value customers are the primary revenue driver and that marketing resources should be disproportionately concentrated on acquiring and retaining the highest-spending customers. In reality, the medium-value segment represents a substantial, stable, and slightly larger revenue contributor that deserves dedicated strategic attention.

6.3 Essential Categories Dominate Spending

The category analysis confirms that over 97% of credit card spending in this portfolio is allocated to non-discretionary essential categories: bills, food, fuel, grocery, and entertainment. Travel, often assumed to be a major driver of premium credit card usage, accounts for only 2.68% of total spending. This finding has two important implications. First, rewards programs and card benefits that are heavily weighted toward travel airport lounges, airline miles, hotel points are misaligned with the actual spending behavior of the overwhelming majority of this portfolio's customers. Second, cashback programs targeting essential spending categories would reach the broadest possible customer base and generate the highest engagement rates.

6.4 Spending Stability Supports Predictive Planning

The monthly trend analysis reveals a high degree of temporal stability in portfolio-level spending, with monthly revenue consistently averaging around Rs. 200M with modest variance. This stability, driven by the dominance of recurring essential spending categories, means that the bank can forecast monthly revenue with reasonable confidence, plan promotional campaigns around predictable spending patterns, and set credit limits and risk provisions based on stable behavioral baselines rather than volatile discretionary spending.

6.5 Basic Card Users Outspend Premium Card Users in Aggregate

The card type analysis reveals that basic card users (Gold and Silver cardholders) collectively generate higher aggregate monthly spending than premium card users (Platinum and Signature cardholders). This is attributable to the larger number of basic card users in the portfolio each individual premium cardholder may spend more per transaction, but the aggregate revenue from the larger basic card user base exceeds the aggregate premium card contribution. Banks should therefore not neglect the basic card segment in favor of premium card acquisition, as the basic card base represents a substantial and stable revenue foundation.

7. Business Impact and Recommendations

7.1 The Business Case for Behavioral Segmentation

The findings of this analysis make a compelling quantitative case for transitioning from uniform customer targeting to behavior-based segmentation. The total portfolio generates Rs. 4.07 billion in revenue across 26,052 transactions. If behavioral segmentation enables even a 5% improvement in targeting efficiency meaning 5% more revenue is generated from the same customer base through better-aligned product offers and rewards the incremental revenue impact would be approximately Rs. 203.5 million. At a 10% improvement, the impact reaches approximately Rs. 239 million. These are conservative estimates based on the assumption that better-aligned incentives increase customer spending frequency and category depth, an assumption well-supported by the behavioral segmentation literature.

7.2 Specific Recommended Actions

Based on the analytical findings, the following specific, actionable recommendations are proposed for the bank's product and marketing strategy teams:

Recommendation 1: Launch Category-Specific Cashback Programs for Medium-Value Customers

Since medium-value customers (30% of the base, 38% of revenue) concentrate their spending on essential categories bills, food, and fuel the bank should design a dedicated credit card product or loyalty tier that offers cashback rewards specifically on these categories. For example, a 2-3% cashback on bills and food transactions would directly incentivize the spending behavior already exhibited by this segment, increasing transaction frequency and customer retention without requiring behavioral change from the customer.

Recommendation 2: Differentiate Premium Card Benefits Toward Travel and Entertainment for High-Value Tier 1 Customers

High-value customers (20% of the base, 37.5% of revenue) in Tier 1 cities spend disproportionately on bills as their dominant category, but individually generate the highest per-transaction amounts. For this segment, the bank should differentiate premium card benefits to include exclusive travel rewards, concierge services, and premium entertainment access benefits that align with the higher discretionary spending capacity of this segment and create a meaningful differentiation from basic card offerings.

Recommendation 3: Design Tier-Differentiated Campaign Strategies

Since Tier 1 and Tier 2 cities exhibit concentrated, category-focused spending behavior while Tier 3 cities show more diversified spending across multiple categories, the bank should not apply a single national promotional strategy. For Tier 1 and Tier 2 cities, focused category-specific campaigns (bills cashback for Tier 1, entertainment offers for Tier 2) will yield higher response rates. For Tier 3 cities, multi-category bundle offers combining food, fuel, and grocery benefits in a single product better match the diverse spending behavior observed in smaller city customers.

Recommendation 4: Invest in Basic Card User Retention Programs

Given that basic card users collectively outspend premium card users in aggregate revenue contribution, the bank should develop targeted retention programs for Gold and Silver cardholders. Offering upgrades to mid-tier benefits, personalized spending insights, and loyalty rewards based on essential spending categories would reinforce this segment's existing spending patterns and increase long-term customer lifetime value.

Recommendation 5: Concentrate Marketing Investment in Top Four Cities

The extreme geographic concentration of revenue in Greater Mumbai, Bengaluru, Ahmedabad, and Delhi means that targeted marketing investment in these four cities will generate significantly higher returns than equal investment spread across all 986 cities in the portfolio. The bank should prioritize customer acquisition campaigns, credit limit upgrades, and retention programs in these four markets, while maintaining a lower-cost, digital-first approach for the long tail of smaller cities.

8. Limitations and Analytical Constraints

This analysis provides substantive and actionable insights into customer spending behavior and segmentation, but several important limitations should be considered when interpreting these findings and applying them to real-world banking decisions.

8.1 Absence of Demographic Data

The dataset does not include demographic attributes such as customer age, income level, credit score, or employment status. As a result, the segmentation model developed in this study is based exclusively on transaction-level behavioral signals—spending amounts, categories, timing, and geography. While behavioral segmentation has been shown to outperform demographic segmentation in many contexts (Rust and Verhoef, 2005), the integration of demographic data would substantially enrich the segmentation model, enabling the bank to identify not only how customers behave but why they behave that way and how their behavior is likely to evolve over their financial life cycle.

8.2 Simulated Dataset

The dataset used in this study is a simulated Kaggle dataset designed to reflect realistic Indian banking transaction patterns rather than actual ICICI Bank or any other bank's proprietary transaction records. While the dataset's structure and value distributions appear realistic and consistent with published information about Indian retail banking markets, the findings cannot be directly extrapolated to any specific bank's actual customer base without validation against real transaction data. The patterns and recommendations generated by this analysis should be treated as directional insights that require empirical validation on actual bank data before implementation.

8.3 Static Snapshot Analysis

The analysis is based on a fixed historical dataset and represents a static snapshot of customer behavior rather than a continuously updated view. Customer spending behavior can shift significantly in response to macroeconomic changes, competitive product launches, regulatory changes, and individual life events. The monthly trend analysis does confirm stability over the analysis period, but this stability cannot be assumed to persist indefinitely. A production-grade implementation of this segmentation model would require integration with real-time transaction feeds and periodic model recalibration to ensure that segment assignments remain accurate as customer behavior evolves.

8.4 Rule-Based Segmentation Limitations

The customer segmentation model implemented in this study uses rule-based thresholds (transaction amount cutoffs) to classify customers into high, medium, and low value segments. While this approach is transparent, interpretable, and computationally efficient, it does not account for the multidimensional nature of customer value. A customer who makes frequent low-value transactions may be more valuable in terms of loyalty and lifetime value than a customer who makes infrequent high-value transactions. Future iterations of this segmentation model should explore algorithmic clustering approaches that can identify more nuanced, multidimensional behavioral segments.

9. Future Work and Enhancements

This project establishes a robust analytical foundation for behavioral customer segmentation in the credit card transaction context. Several enhancements would substantially increase the model's accuracy, real-world applicability, and commercial value.

9.1 Clustering Algorithms for Data-Driven Segmentation

Replacing the current rule-based segmentation approach with machine learning clustering algorithms such as K-Means or DBSCAN would enable the model to identify customer segments based on natural patterns in the data rather than manually defined thresholds. For example, K-Means clustering applied to a feature set including average transaction amount, primary spending category, transaction frequency, geographic tier, and card type would likely reveal more nuanced segments such as frequent small-ticket essential spenders, occasional large-ticket discretionary spenders, or geographically mobile multi-city customers that the current three-tier model cannot distinguish.

9.2 Real-Time Transaction Integration

Integrating real-time transaction data streams would transform this analysis from a static historical snapshot into a dynamic, continuously updated monitoring system. With real-time data, the bank could detect behavioral shifts as they occur for example, identifying customers who are increasing their travel spending and immediately presenting them with relevant travel rewards offers rather than waiting for monthly or quarterly analysis cycles. This real-time capability would significantly

increase the commercial value of the segmentation model.

9.3 Predictive Modeling

Developing predictive models on top of the current behavioral segmentation framework would enable the bank to forecast customer value trajectory rather than simply measuring current value. For example, a gradient boosting model trained on historical transaction patterns could predict which low-value or medium-value customers are likely to transition to the high-value segment within the next six months, enabling proactive targeting with upgrade offers before the behavioral shift occurs rather than after. Similarly, churn prediction models could identify customers whose spending patterns suggest declining engagement, enabling retention interventions before the customer becomes inactive.

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